Statement of Research Interests

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Text is the interface between people and language models, but models also use it to reason, access knowledge, and communicate with other agents, which is slow and costly. My research asks how much of this work can happen directly in learned latent representations. I want to use reinforcement learning to train these representations and understand when they help. I am especially interested in agents that can retain useful skills and knowledge, then build on them through further learning. This raises questions about how models learn from errors and how we evaluate what they learn.

In KV-Skill, I developed the operator architecture and reward-learning formulation for reusable skills in a frozen model. A collaborator first proposed optimizing KV caches to store skills, but differences in attention architectures made this difficult to apply consistently across models, so I wanted a design that could be trained for different architectures at low resource cost. I compared interventions at the KV cache, after rotary positional encoding, and in the residual stream. The residual stream worked best in our experiments. I then designed and tested several operator variants inspired by DeltaNet. Our final operators were best or tied-best in seven of eight matched-budget settings against soft prefixes, prefix tuning, and LoRA. The reward-learning formulation supports learning from an existing text skill or from scratch. A recent paper from Prof. Lei Chen's group at HKUST called KV-Skill "a striking demonstration" and built directly on it as its main baseline.

I am now exploring how latent representations can give models access to external knowledge. In one direction, a small model prefills knowledge into latent form that a much larger model reads directly. We have promising preliminary results.

My experiments on latent reasoning showed me why moving beyond text is difficult. I tested whether redundant chain-of-thought steps could move into latent space under RL post-training. A routing signal identified filler steps offline but provided little useful guidance during RL. Removing the discrete token bottleneck also weakened error correction. These results made me more interested in credit assignment. How can training identify which latent computations help, and which lead the model astray? They also made me question what useful structure text provides during reasoning.

Credit assignment is also what I worked on in SCRIBE, a literature-review agent I designed and built that is now deployed in GLKB.org. I translated our evaluation framework into rubric-grounded rewards at stage boundaries, which gave me experience rewarding intermediate decisions that a single trajectory reward can miss. The resulting paper is under review at ICLR 2027 under the title What Can a Leaderboard Certify? I am also experienced with veRL, OpenRLHF, DeepSpeed, and FSDP.

Future Research Interests

Latent reasoning with RL. I want to study how post-training shapes latent computation. A central question is how to reduce the need for explicit reasoning tokens while preserving a model's ability to recognize and recover from mistakes.

Latent credit assignment. Can intermediate latent states tell us which computations contributed to an outcome? I want to test whether those signals can improve credit assignment during RL. This would make representations a source of feedback for learning.

Latent communication. Multi-agent systems spend heavy compute writing and rereading text, and my work on skills and knowledge access gives me a starting point for studying latent communication between them. I want to understand what learned interfaces preserve and how reliably another model can use that information.

Self-improving agents. In the longer term, I want to build agents whose capabilities accumulate in reusable latent modules. I am interested in how to evaluate, update, and remove those modules as the agent learns. We also need ways to track how each change affects its behavior.